

STAT6018 Research Frontiers in Data Science

Topic I: Statistical methods for analyzing complex survival data

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- 1 Chapter 1: Semiparametric transformation models for censored data
 - Transformation models for counting processes
 - Transformation models with random effects for recurrent events
 - Joint analysis of recurrent and terminal events
 - Frailty transformation models for multivariate survival data

Course Logistics

Lectures:

- Weeks 1–3
- Mainly discuss statistical papers on survival analysis

Presentations:

- Week 4
- Alphabetical order by last name
- Attendance is required
- Format: 15-min presentation + 5-min Q&A
- Topic: any paper/project related to survival analysis

Censored Data

Univariate survival data: time to the occurrence of a given event/failure

- Time to death
- Time to the occurrence of a disease

Multivariate survival data: times to several events/failures

- Recurrent events: repetitions of a phenomenon (e.g., illness)
 - ▶ Tumor recurrences
 - ▶ Infection episodes
- Multiple types of events: combination of multiple types of phenomena
 - ▶ Ordered events, such as HIV-infection → AIDS → death
 - ▶ Unordered events, such as diseases in several organ systems (cardiovascular disease, cancer, Alzheimer's disease, etc.)

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Reference

-  Zeng, D., & Lin, D. Y. (2006). Efficient estimation of semiparametric transformation models for counting processes. *Biometrika*, 93(3), 627-640.

Counting processes

- Counting process is a continuous-time stochastic process $\{N(t) : t \geq 0\}$ with $N(0) = 0$, whose sample paths are step functions with jumps of size 1 only.
- In survival analysis without censoring, $N(t)$ records the number of events that have occurred by time t .
- For univariate survival data, $N(t)$ takes a single jump at the survival time.
- For recurrent events data, $N(t)$ takes jumps at all recurrent event times.

Intensity function

Notation:

- $N^*(t)$: counting process recording the number of events by time t
- $X(t)$: potentially time-dependent covariates
- $\mathcal{F}_t = \{N^*(s), X(s) : 0 \leq s \leq t\}$: history up to time t
- $dN^*(t)$: increment of N^* (i.e., number of events) over $[t, t + dt)$

Intensity function:

$$\lambda(t|X) = \lim_{dt \downarrow 0} \frac{1}{dt} E\{dN^*(t) \mid \mathcal{F}_{t-}\}$$

Cumulative intensity function:

$$\Lambda(t|X) = \int_0^t \lambda(s|X) ds$$

Proportional intensity model

Proportional intensity (PI) model:

$$\Lambda(t|X) = \int_0^t Y^*(s) \exp \{ \beta^T X(s) \} d\Lambda(s)$$

- $Y^*(t)$: indicator process
 - ▶ $Y^*(t) = I(T \geq t)$ for univariate survival data
 - ▶ $Y^*(t) \equiv 1$ for recurrent events data
- $\Lambda(t)$: unknown cumulative baseline intensity function
- β : unknown regression parameters

A large-sample theory for this model based on [maximum partial likelihood estimation](#) has been established via the [counting-process martingale theory](#)¹.

¹ Andersen, P. K., & Gill, R. D. (1982). Cox's regression model for counting processes: a large sample study. *The Annals of Statistics*, 10, 1100-1120.

Discussion about PI model

- For univariate survival data, the PI model reduces to the Cox proportional hazards (PH) model.
- The proportional hazards assumption may be violated in certain applications, especially in long-term studies.
- For example, the initial effect of a treatment may disappear with time, such that the hazard ratio converges to 1 as $t \rightarrow \infty$.
- A useful alternative is the **proportional odds (PO) model**²:

$$\frac{\Pr(T \leq t|X)}{\Pr(T > t|X)} = g(t) \exp \{ \beta^T X(t) \},$$

which constrains the hazard ratio to converge to 1 as $t \rightarrow \infty$.

²Bennett, S. (1983). Analysis of survival data by the proportional odds model. *Statistics in medicine*, 2(2), 273-277.

Semiparametric transformation models

The PH/PI and PO models belong to the broad class of **semiparametric transformation models** for general counting processes:

$$\Lambda(t|X) = G \left[\int_0^t Y^*(s) \exp \{ \beta^T X(s) \} d\Lambda(s) \right] \quad (1)$$

- $G(\cdot)$: strictly increasing transformation function
 - ▶ $G(x) = x \Rightarrow$ PH/PI model
 - ▶ $G(x) = \log(1 + x) \Rightarrow$ PO model
- $\Lambda(t)$: arbitrary increasing function

Common choices of transformations

Box-Cox transformations:

$$G(x) = \rho^{-1} \{(1+x)^\rho - 1\} \quad (\rho \geq 0)$$

Logarithmic transformations:

$$G(x) = r^{-1} \log(1+rx) \quad (r \geq 0)$$

- $\rho = 1$ or $r = 0 \Rightarrow$ PH/PI model
- $\rho = 0$ or $r = 1 \Rightarrow$ PO model

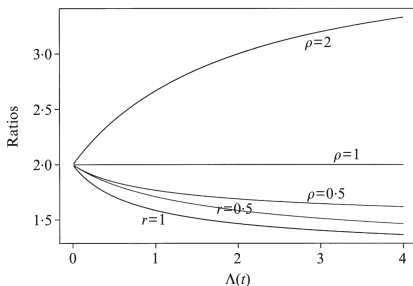


Figure 1: Plots of $\Lambda(t|X=x)/\Lambda(t|X=0)$ against $\Lambda(t)$ with $e^{\beta^T x} = 2$

Censored counting processes

Notation:

- C : censoring time
- $N(t) = N^*(t \wedge C)$: counting process recording the number of events observed by time t
- $Y(t) = Y^*(t)I(C \geq t)$: at-risk indicator process
- τ : study end time

Independent censoring assumption: $N^*(t) \perp\!\!\!\perp C$ conditional on $X(t)$

Observed data from n random samples:

$$\left\{ N_i(t), Y_i(t), X_i(t) : t \in [0, \tau] \right\} \quad \text{for } i = 1, \dots, n$$

Likelihood

Define $\lambda(t) = \Lambda'(t)$. Under model (1), the intensity function for $N_i(t)$ is

$$\lambda(t|X_i) = Y_i(t)e^{\beta^\top X_i(t)}\lambda(t)G' \left\{ \int_0^t Y_i(s)e^{\beta^\top X_i(s)}d\Lambda(s) \right\}.$$

Thus, the likelihood function is

$$\begin{aligned} L_n(\beta, \Lambda) &= \prod_{i=1}^n \prod_{t \in [0, \tau]} \lambda(t|X_i)^{dN_i(t)} \exp \{ -\Lambda(\tau|X_i) \} \\ &= \prod_{i=1}^n \prod_{t \in [0, \tau]} \left[e^{\beta^\top X_i(t)}\lambda(t)G' \left\{ \int_0^t Y_i(s)e^{\beta^\top X_i(s)}d\Lambda(s) \right\} \right]^{dN_i(t)} \\ &\quad \times \exp \left[-G \left\{ \int_0^\tau Y_i(s)e^{\beta^\top X_i(s)}d\Lambda(s) \right\} \right]. \end{aligned}$$

Likelihood (cont.)

And the log-likelihood function is

$$\begin{aligned}\ell_n(\beta, \Lambda) = \sum_{i=1}^n & \left(\int_0^\tau \{ \beta^\top X_i(t) + \log \lambda(t) \} dN_i(t) \right. \\ & + \int_0^\tau \log G' \left\{ \int_0^t Y_i(s) e^{\beta^\top X_i(s)} d\Lambda(s) \right\} dN_i(t) \\ & \left. - G \left\{ \int_0^\tau Y_i(s) e^{\beta^\top X_i(s)} d\Lambda(s) \right\} \right).\end{aligned}$$

We maximize the log-likelihood over β and Λ .

NPMLE

- We adopt the **nonparametric maximum likelihood estimation (NPMLE)** approach, where Λ is restricted to be a step function with non-negative jumps at all the observed event times, denoted by $t_1 < t_2 < \dots < t_m$.
- The log-likelihood function under NPMLE becomes

$$\begin{aligned}\ell_n(\beta, \Lambda) = & \sum_{i=1}^n \left(\int_0^\tau \{ \beta^\top X_i(t) + \log \Lambda\{t\} \} dN_i(t) \right. \\ & + \int_0^\tau \log G' \left\{ \sum_{k:t_k \leq t} e^{\beta^\top X_i(t_k)} \Lambda\{t_k\} \right\} dN_i(t) \\ & \left. - G \left\{ \sum_{k:t_k \leq C_i} e^{\beta^\top X_i(t_k)} \Lambda\{t_k\} \right\} \right),\end{aligned}$$

where $\Lambda\{t\}$ denotes the jump size of Λ at time t .

- The estimators of β and $\Lambda\{t_k\}$ ($k = 1, \dots, m$) are obtained via the quasi-Newton method.

Variance estimation

To estimate the limiting covariance function of $\sqrt{n}(\hat{\beta} - \beta_0, \hat{\Lambda} - \Lambda_0)$, it suffices to obtain a variance estimator for the linear functional

$$\sqrt{n} \int_0^\tau w(t) d\{\hat{\Lambda}(t) - \Lambda_0(t)\} + \sqrt{n} b^T (\hat{\beta} - \beta_0),$$

where $w(\cdot) \in \text{BV}([0, \tau])$ and $b \in \mathbb{R}^p$.

We can treat β and $\Lambda\{t_k\}$'s as the parameters and estimate their limiting covariance matrix by the inverse of the observed information matrix $n\mathcal{I}_n$.

Since $\sqrt{n} \int_0^\tau w(t) d\{\hat{\Lambda}(t) - \Lambda_0(t)\} + \sqrt{n} b^T (\hat{\beta} - \beta_0)$ is linear with all parameter estimates, its limiting variance V can be estimated by

$$\hat{V} = (W^T \quad b^T) \mathcal{I}_n^{-1} \begin{pmatrix} W \\ b \end{pmatrix},$$

where W is the vector of $w(\cdot)$ evaluated at all observed event times.

Asymptotic properties

Let $(\hat{\beta}, \hat{\Lambda})$ and (β_0, Λ_0) denote the nonparametric maximum likelihood estimates and the true values of (β, Λ) , respectively. We have:

Consistency: $\|\hat{\beta} - \beta_0\| + \sup_{t \in [0, \tau]} |\hat{\Lambda} - \Lambda_0| \xrightarrow{a.s.} 0$.

Asymptotic normality: $\sqrt{n}(\hat{\beta} - \beta_0, \hat{\Lambda} - \Lambda_0)$ converges weakly to a mean-zero Gaussian process.

Semiparametric efficiency: The limiting covariance matrix of $\hat{\beta}$ attains the semiparametric efficiency bound.

Consistency of variance estimators: $\hat{V} \xrightarrow{a.s.} V$.

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Motivation

- Recall the proportional intensity model for recurrent events

$$\lambda(t|X) = \lambda(t) \exp \{ \beta^T X(t) \}$$

- Under the above model, the occurrence of an event is independent of any earlier events of the same subjects, which may not hold true in practice.
- For example, people who had a previous COVID-19 infection tend to have a lower risk of reinfection, while people who develop tumors more quickly than others tend to experience tumor recurrence more quickly.
- We could let $X(t)$ include the past event history, but this is not ideal since modeling the within-subject correlation through time-dependent covariates is very difficult.

PI model with frailty

- A useful approach to accommodating the dependence of the recurrent event times within the same subject is to incorporate a random effect (or frailty) into the model:

$$\lambda(t|X) = \xi \lambda(t) \exp \{ \beta^T X(t) \}$$

- The frailty ξ may capture the within-subject correlation and is usually assumed to follow the Gamma distribution.
- However, gamma frailty induces a very restrictive form of dependence.

Transformation models with random effects

We specify that the cumulative intensity function of $N^*(t)$ takes the form

$$\Lambda(t|X, Z, b) = G \left[\int_0^t \exp \{ \beta^T X(s) + \textcolor{red}{b}^T \textcolor{red}{Z}(s) \} d\Lambda(s) \right]$$

- b : subject-specific random effects with mean 0 and density function $\phi(b; \gamma)$, used to capture the within-subject correlation
- $X(t)$ and $Z(t)$: potentially time-dependent covariates, may include covariates derived from the event history before time t
- b is usually assumed to follow a mean-zero multivariate normal distribution.

Recurrent events data

Observed data from n random samples:

$$\left\{ N_i(t), Y_i(t), X_i(t), Z_i(t) : t \in [0, \tau] \right\} \quad \text{for } i = 1, \dots, n$$

- $N_i(t) = N_i^*(t \wedge C_i)$
- $Y_i(t) = I(C_i \geq t)$

Independent censoring assumption: The conditional density of C at t given $\{N^*(s), X(s), Z(s) : s \in [0, \tau]\}$ and b depends only on $\{X(s), Z(s) : s \leq t\}$ and is noninformative about (β, γ, Λ) .

Noninformative covariate processes assumption: The conditional distribution of $\{X(t), Z(t)\}$ given $\{N(s), Y(s), X(s), Z(s) : s < t\}$ is noninformative about (β, γ, Λ) .

Likelihood and NPMLE

Let $\theta = (\beta^\top, \gamma^\top)^\top$. The likelihood function under the preceding two assumptions is

$$L_n(\theta, \Lambda) = \prod_{i=1}^n \int_{b_i} \prod_{t \in [0, \tau]} \left[\lambda(t) e^{\beta^\top X_i(t) + b_i^\top Z_i(t)} G' \left\{ \int_0^t Y_i(s) e^{\beta^\top X_i(s) + b_i^\top Z_i(s)} d\Lambda(s) \right\} \right]^{dN_i(t)} \\ \times \exp \left[-G \left\{ \int_0^\tau Y_i(s) e^{\beta^\top X_i(s) + b_i^\top Z_i(s)} d\Lambda(s) \right\} \right] \phi(b_i; \gamma) db_i$$

NPMLE: Λ is treated as a step function with non-negative jumps at all the observed event times.

EM algorithm

The estimators can be computed via an EM algorithm, treating the random effects b_i as missing data.

The complete-data log-likelihood function is

$$\begin{aligned}\ell_c(\theta, \Lambda) = & \sum_{i=1}^n \left(\int_0^\tau \{ \beta^\top X_i(t) + b_i^\top Z_i(t) + \log \Lambda\{t\} \} dN_i(t) \right. \\ & + \int_0^\tau \log G' \left\{ \int_0^t Y_i(s) e^{\beta^\top X_i(s) + b_i^\top Z_i(s)} d\Lambda(s) \right\} dN_i(t) \\ & \left. - G \left\{ \int_0^\tau Y_i(s) e^{\beta^\top X_i(s) + b_i^\top Z_i(s)} d\Lambda(s) \right\} + \log \phi(b_i; \gamma) \right).\end{aligned}$$

E-step

Let $\widehat{E}(\cdot)$ denote the conditional expectation given the observed data.

In the E-step, we compute $\widehat{E}\{H(b_i)\}$ for some function $H(\cdot)$ based on the posterior density of b_i , which is proportional to

$$\prod_{i=1}^n \prod_{t \in [0, \tau]} \left[\lambda(t) e^{\beta^\top X_i(t) + b_i^\top Z_i(t)} G' \left\{ \int_0^t Y_i(s) e^{\beta^\top X_i(s) + b_i^\top Z_i(s)} d\Lambda(s) \right\} \right]^{dN_i(t)} \\ \times \exp \left[-G \left\{ \int_0^\tau Y_i(s) e^{\beta^\top X_i(s) + b_i^\top Z_i(s)} d\Lambda(s) \right\} \right] \phi(b_i; \gamma)$$

The integral over b_i in $\widehat{E}\{H(b_i)\}$ can be approximated by Gauss–Hermite quadrature.

M-step

In the M-step, we maximize the objective function

$$\begin{aligned} M(\theta, \Lambda) = & \sum_{i=1}^n \left(\int_0^\tau \{ \beta^\top X_i(t) + \log \Lambda\{t\} \} dN_i(t) \right. \\ & + \int_0^\tau \hat{E} \left[b_i^\top Z_i(t) + \log G' \left\{ \int_0^t Y_i(s) e^{\beta^\top X_i(s) + b_i^\top Z_i(s)} d\Lambda(s) \right\} \right] dN_i(t) \\ & \left. - \hat{E} \left[G \left\{ \int_0^\tau Y_i(s) e^{\beta^\top X_i(s) + b_i^\top Z_i(s)} d\Lambda(s) \right\} \right] + \hat{E} \{ \log \phi(b_i; \gamma) \} \right). \end{aligned}$$

We update γ by maximizing $\sum_{i=1}^n \hat{E} \{ \log \phi(b_i; \gamma) \}$.

M-step (cont.)

To update β and Λ , define $F(t) = \Lambda(t)/\Lambda(\tau)$. We expand β to $[\log \Lambda(\tau), \beta]$ and expand $X_i(t)$ to $[1, X_i(t)]$. For simplicity, we still denote the expanded terms by β and $X_i(t)$.

Then the objective function to be maximized is equivalent to

$$\begin{aligned}\tilde{M}(\beta, F) = & \sum_{i=1}^n \left(\int_0^\tau \{ \beta^\top X_i(t) + \log F\{t\} \} dN_i(t) \right. \\ & + \int_0^\tau \hat{E} \left[b_i^\top Z_i(t) + \log G' \left\{ \int_0^t Y_i(s) e^{\beta^\top X_i(s) + b_i^\top Z_i(s)} dF(s) \right\} \right] dN_i(t) \\ & \left. - \hat{E} \left[G \left\{ \int_0^\tau Y_i(s) e^{\beta^\top X_i(s) + b_i^\top Z_i(s)} dF(s) \right\} \right] \right),\end{aligned}$$

with the constraint that $\sum_{i=1}^n \int_0^\tau F\{t\} dN_i(t) = 1$ (by NPMLE).

M-step (cont.)

Notation:

- T_{ij} : j th event time of the i th subject ($i = 1, \dots, n$ and $j = 1, \dots, n_i$)
- $t_1 < t_2 < \dots < t_m$: sorted sequence of all distinct values of T_{ij}
- $f_k = F\{t_k\}$, for $k = 1, \dots, m$
- μ : Lagrange multiplier

The objective function can be written as

$$\begin{aligned}\tilde{M}(\beta, F) = & \sum_{k=1}^m \log(f_k) + \sum_{i=1}^n \left(\sum_{j=1}^{n_i} \beta^\top X_i(T_{ij}) \right. \\ & + \sum_{j=1}^{n_i} \hat{E} \left[b_i^\top Z_i(T_{ij}) + \log G' \left\{ \sum_{k: t_k \leq T_{ij}} e^{\beta^\top X_i(t_k) + b_i^\top Z_i(t_k)} f_k \right\} \right] \\ & \left. - \hat{E} \left[G \left\{ \sum_{k: t_k \leq C_i} e^{\beta^\top X_i(t_k) + b_i^\top Z_i(t_k)} f_k \right\} \right] \right) - \mu \left(\sum_{k=1}^m f_k - 1 \right).\end{aligned}$$

M-step (cont.)

We then solve the score equations for β and $(f_1, \dots, f_m)$:

$$0 = \sum_{i=1}^n \left(\sum_{j=1}^{n_i} X_i(T_{ij}) \right. \\ \left. + \sum_{j=1}^{n_i} \hat{E} \left[\frac{G'' \{ \sum_{k: t_k \leq T_{ij}} e^{\beta^T X_i(t_k) + b_i^T Z_i(t_k)} f_k \}}{G' \{ \sum_{k: t_k \leq T_{ij}} e^{\beta^T X_i(t_k) + b_i^T Z_i(t_k)} f_k \}} \times \sum_{k: t_k \leq T_{ij}} e^{\beta^T X_i(t_k) + b_i^T Z_i(t_k)} X_i(t_k) f_k \right] \right. \\ \left. - \hat{E} \left[G' \left\{ \sum_{k: t_k \leq C_i} e^{\beta^T X_i(t_k) + b_i^T Z_i(t_k)} f_k \right\} \times \sum_{k: t_k \leq C_i} e^{\beta^T X_i(t_k) + b_i^T Z_i(t_k)} X_i(t_k) f_k \right] \right).$$

and

$$\mu = \frac{1}{f_k} + \sum_{i=1}^n \left(\sum_{j=1}^{n_i} \hat{E} \left[\frac{G'' \{ \sum_{l: t_l \leq T_{ij}} e^{\beta^T X_i(t_l) + b_i^T Z_i(t_l)} f_l \}}{G' \{ \sum_{l: t_l \leq T_{ij}} e^{\beta^T X_i(t_l) + b_i^T Z_i(t_l)} f_l \}} \times I(t_k \leq T_{ij}) e^{\beta^T X_i(t_k) + b_i^T Z_i(t_k)} \right] \right. \\ \left. - \hat{E} \left[G' \left\{ \sum_{l: t_l \leq C_i} e^{\beta^T X_i(t_l) + b_i^T Z_i(t_l)} f_l \right\} \times I(t_k \leq C_i) e^{\beta^T X_i(t_k) + b_i^T Z_i(t_k)} \right] \right)$$

Recursive formula

When $X(t)$ and $Z(t)$ are both time-independent, it is easy to observe that the second equation provides a recursive formula for calculating $(f_1, \dots, f_m)$:

$$\frac{1}{f_{k+1}} = \frac{1}{f_k} + \sum_{i=1}^n \left(\sum_{j=1}^{n_i} \hat{E} \left[\frac{G'' \{ e^{\beta^\top X_i + b_i^\top Z_i} F(t_k) \}}{G' \{ e^{\beta^\top X_i + b_i^\top Z_i} F(t_k) \}} \times I(T_{ij} = t_k) e^{\beta^\top X_i + b_i^\top Z_i} \right] \right. \\ \left. - \hat{E} \left[G' \{ e^{\beta^\top X_i + b_i^\top Z_i} F(t_k) \} \times I(t_k \leq C_i < t_{k+1}) e^{\beta^\top X_i + b_i^\top Z_i} \right] \right)$$

Write f_k as $f_k(f_1, \beta)$. We can solve (f_1, β) via the Newton-Raphson method, where the derivatives of f_k w.r.t. f_1 and β are calculated based on the above recursive formula, with initial values $\partial f_1 / \partial f_1 = 1$ and $\partial f_1 / \partial \beta = 0$.

This addresses the issue of high-dimensional parameters in NPMLE.

Variance estimation

As in the previous paper, the limiting variances of $(\hat{\beta}, \hat{\Lambda})$ can be consistently estimated by the inverse of the observed information matrix $n\mathcal{I}_n$.

By Louis' formula³, $n\mathcal{I}_n$ can be calculated within the EM algorithm by

$$-\sum_{i=1}^n \hat{E} \{ \nabla^2 \ell_i(b_i; \theta, \Lambda) \} - \sum_{i=1}^n \left[\hat{E} \{ \nabla \ell_i(b_i; \theta, \Lambda)^{\otimes 2} \} - \hat{E} \{ \nabla \ell_i(b_i; \theta, \Lambda) \}^{\otimes 2} \right],$$

where ℓ_i is the i th subject's contribution to the complete-data log-likelihood function, and $\nabla \ell_i$ denotes the gradient of ℓ_i w.r.t. β and $\Lambda\{t_k\}$'s.

³Louis, T. A. (1982). Finding the observed information matrix when using the EM algorithm. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 44(2), 226-233.

Asymptotic properties under known G

Let $(\hat{\theta}, \hat{\Lambda})$ and (θ_0, Λ_0) denote the nonparametric maximum likelihood estimates and the true values of (θ, Λ) , respectively.

When the transformation $G(\cdot)$ is completely specified, we have:

Consistency: $\|\hat{\theta} - \theta_0\| + \sup_{t \in [0, \tau]} |\hat{\Lambda} - \Lambda_0| \xrightarrow{a.s.} 0$.

Asymptotic normality: $\sqrt{n}(\hat{\theta} - \theta_0, \hat{\Lambda} - \Lambda_0)$ converges weakly to a mean-zero Gaussian process.

Semiparametric efficiency: The limiting covariance matrix of $\hat{\theta}$ attains the semiparametric efficiency bound.

Asymptotic properties under unknown G

- When the transformation $G(\cdot)$ belongs to a one-parameter family $\{G_\eta : \eta \in (a_0, b_0)\}$, η is another unknown parameter.
- Write $\theta = (\beta^\top, \gamma^\top, \eta)^\top$. With some additional conditions, all the asymptotic properties on the previous slide still hold.
 - ▶ Linear independence of covariates at time 0
 - ▶ Smoothness conditions for G_η w.r.t. η
- The Box–Cox and logarithmic transformations introduced before satisfy those additional conditions, so their parameters (ρ or r) can also be estimated from the data.

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-  Zeng, D., & Lin, D. Y. (2009). Semiparametric transformation models with random effects for joint analysis of recurrent and terminal events. *Biometrics*, 65(3), 746-752.

Motivation

- In practice, recurrent event times are subject to censoring. Most of the existing methods require independent censoring.
- This is OK if censoring is caused by the end of the study or random loss to follow-up.
- In many medical studies, however, recurrent events may be terminated by the subject's withdrawal from the study due to deteriorating health or the subject's death.
- In those cases, the censoring time is likely correlated with the recurrent event times, and existing methods may yield misleading results.
- To address the dependent censoring issue, we consider joint analysis of recurrent and terminal events through shared random effects models.

Joint transformation models

Submodel for recurrent event process $N^*(t)$:

$$\Lambda_R(t|X, Z, b) = H \left[\int_0^t \exp \{ \alpha^\top X(s) + b^\top Z(s) \} dA(s) \right]$$

Submodel for terminal event time T :

$$\Lambda_T(t|X, Z, b) = G \left[\int_0^t \exp \{ \beta^\top X(s) + b^\top (\gamma \circ Z(s)) \} d\Lambda(s) \right]$$

- $H(\cdot)$ and $G(\cdot)$: transformation functions
- α , β , and γ : unknown regression parameters
- $X(t)$ and $Z(t)$: potentially time-dependent covariates, $Z(t)$ contains 1
- $\gamma \circ Z(s)$: component-wise product of γ and $Z(s)$
- b : shared random effects, with mean 0 and density function $\phi(b; \eta)$

Joint transformation models (cont.)

Submodel for recurrent event process $N^*(t)$:

$$\Lambda_R(t|X, Z, b) = H \left[\int_0^t \exp \{ \alpha^\top X(s) + \textcolor{red}{b}^\top Z(s) \} dA(s) \right]$$

Submodel for terminal event time T :

$$\Lambda_T(t|X, Z, b) = G \left[\int_0^t \exp \{ \beta^\top X(s) + \textcolor{red}{b}^\top (\gamma \circ Z(s)) \} d\Lambda(s) \right]$$

- The variance of b characterizes the dependence among recurrent event times.
- γ characterizes the dependence between recurrent and terminal events attributed to the unobserved random effects. $\gamma = 0$ implies that the dependence can be fully explained by the covariates.

Data and assumption

Data: $\{Y_i, \Delta_i, N_i^*(t), X_i(t), Z_i(t) : t \leq Y_i\}$ ($i = 1, \dots, n$)

- $Y_i = \min(T_i, C_i)$
- $\Delta_i = I(T_i \leq C_i)$
- C_i : censoring time

Independent censoring assumption: $C_i \perp\!\!\!\perp (N_i^*, T_i, b_i)$ conditional on the covariates X_i and Z_i

Conditional independence: $N_i^* \perp\!\!\!\perp T_i$ conditional on b_i, X_i , and Z_i

Likelihood

Let $a(t) = A'(t)$, $\lambda(t) = \Lambda'(t)$, and $R_i(t) = I(Y_i \geq t)$. The observed-data likelihood function concerning $(\alpha, \beta, \gamma, \eta, A, \Lambda)$ is

$$\begin{aligned} & \prod_{i=1}^n \int_{b_i} \left[\prod_t \left\{ a(t) e^{\alpha^\top X_i(t) + b_i^\top Z_i(t)} H' \left(\int_0^t e^{\alpha^\top X_i(s) + b_i^\top Z_i(s)} dA(s) \right) \right\}^{R_i(t) dN_i^*(t)} \right. \\ & \quad \times \exp \left\{ -H \left(\int_0^{Y_i} e^{\alpha^\top X_i(t) + b_i^\top Z_i(t)} dA(t) \right) \right\} \Big] \\ & \quad \times \left[\left\{ \lambda(Y_i) e^{\beta^\top X_i(Y_i) + b_i^\top (\gamma \circ Z_i(Y_i))} G' \left(\int_0^{Y_i} e^{\beta^\top X_i(t) + b_i^\top (\gamma \circ Z_i(t))} d\Lambda(t) \right) \right\}^{\Delta_i} \right. \\ & \quad \times \exp \left\{ -G \left(\int_0^{Y_i} e^{\beta^\top X_i(t) + b_i^\top (\gamma \circ Z_i(t))} d\Lambda(t) \right) \right\} \Big] \phi(b_i; \eta) db_i \end{aligned}$$

NPMLE

We consider A as a step function with jumps only at the observed recurrent event times, and consider Λ as a step function with jumps only at the observed terminal event times.

Thus, we maximize the following modified log-likelihood function over $(\alpha, \beta, \gamma, \eta)$ and the jump sizes of A and Λ :

$$\begin{aligned} \sum_{i=1}^n \log \int_{b_i} \left[\prod_t \left\{ A\{t\} e^{\alpha^\top X_i(t) + b_i^\top Z_i(t)} H' \left(\int_0^t e^{\alpha^\top X_i(s) + b_i^\top Z_i(s)} dA(s) \right) \right\}^{R_i(t) dN_i^*(t)} \right. \\ \left. \times \exp \left\{ -H \left(\int_0^{Y_i} e^{\alpha^\top X_i(t) + b_i^\top Z_i(t)} dA(t) \right) \right\} \right] \\ \times \left[\left\{ \Lambda \{Y_i\} e^{\beta^\top X_i(Y_i) + b_i^\top (\gamma \circ Z_i(Y_i))} G' \left(\int_0^{Y_i} e^{\beta^\top X_i(t) + b_i^\top (\gamma \circ Z_i(t))} d\Lambda(t) \right) \right\}^{\Delta_i} \right. \\ \left. \times \exp \left\{ -G \left(\int_0^{Y_i} e^{\beta^\top X_i(t) + b_i^\top (\gamma \circ Z_i(t))} d\Lambda(t) \right) \right\} \right] \phi(b_i; \eta) db_i \end{aligned}$$

Computing algorithm

- We may use quasi-Newton or other optimization algorithms to obtain the NPMLEs.
- Alternatively, we can use an EM algorithm for computation, with the subject-specific random effects b_i treated as missing data.
- In the M-step, the maximization is taken over only a small set of parameters, thanks to some recursive formulae among the jump sizes of A and Λ .

Asymptotic properties

Let $\theta = (\alpha^\top, \beta^\top, \gamma^\top, \eta^\top)^\top$ denote the set of all finite-dimensional parameters. We have:

Consistency: $\|\hat{\theta} - \theta_0\| + \sup_{t \in [0, \tau]} |\hat{A} - A_0| + \sup_{t \in [0, \tau]} |\hat{\Lambda} - \Lambda_0| \xrightarrow{a.s.} 0$.

Asymptotic normality: $\sqrt{n}(\hat{\theta} - \theta_0, \hat{A} - A_0, \hat{\Lambda} - \Lambda_0)$ converges weakly to a mean-zero Gaussian process.

Semiparametric efficiency: The limiting covariance matrix of $\hat{\theta}$ attains the semiparametric efficiency bound.

The limiting variances and covariances can be consistently estimated by inverting the observed information matrix for all parameters, including θ and the jump sizes of A and Λ . The observed information matrix can be calculated by Louis' formula.

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- 1 Chapter 1: Semiparametric transformation models for censored data
 - Transformation models for counting processes
 - Transformation models with random effects for recurrent events
 - Joint analysis of recurrent and terminal events
 - Frailty transformation models for multivariate survival data

Reference

-  Zeng, D., Chen, Q., & Ibrahim, J. G. (2009). Gamma frailty transformation models for multivariate survival times. *Biometrika*, 96(2), 277-291.

Multivariate failure time data

- Multivariate failure time data arise when each study subject can experience several events.
- It is interesting to determine risk factors that are predictive for some or all of the failures.
- For example, in COVID-19 vaccine trials, investigators want to access the efficacy of a vaccine against infection, hospitalization, and death.
- Like recurrent events data, multivariate failure times from the same subject are potentially correlated. Ignoring such correlation may lead to biased inference.

Gamma frailty transformation models

Let T_k denote the failure time of the k th event type ($k = 1, \dots, K$). We specify the following gamma frailty transformation model:

$$\Lambda_k(t|X, \xi) = \xi G_k \left\{ \Lambda_k(t) e^{\beta_k^T X} \right\} \quad (2)$$

- $\xi \sim \text{Gamma}(\gamma^{-1}, \gamma)$: captures the within-subject correlation
- $G_k(\cdot)$: type-specific transformation function
- $\Lambda_k(t)$: unspecified type-specific increasing function
- β_k : type-specific regression parameters

Gamma frailty transformation models (cont.)

- Under model (2), the marginal cumulative hazard function for T_k is

$$\Lambda_{T_k}(t) = \gamma^{-1} \log \left[1 + \gamma G_k \left\{ \Lambda_k(t) e^{\beta_k^T X} \right\} \right]$$

- The above marginal distribution is equivalent to another linear transformation model:

$$\log \Lambda_k(T_k) = -\beta_k^T X + \epsilon_k,$$

with ϵ_k following the distribution $\log G_k^{-1}[\gamma^{-1}\{\text{Unif}(0, 1)^{-\gamma} - 1\}]$.

- The dependence among failure times can be evaluated through γ . We allow $\gamma = 0$, which corresponds to the scenario with independent failure times.

Reparameterization

Let τ denote the study end time. We define $F_k(t) = \Lambda_k(t)/\Lambda_k(\tau)$ and $\alpha_k = \log \Lambda_k(\tau)$. Model (2) can be rewritten as

$$\Lambda_k(t|X, \xi) = \xi G_k \left\{ F_k(t) e^{\alpha_k + \beta_k^\top X} \right\} \quad (3)$$

Clearly, $F_k(\cdot)$ is a distribution function in $[0, \tau]$, with $F_k(0) = 0$ and $F_k(\tau) = 1$.

Under some mild conditions on the true parameter values, the transformation functions, and the censoring distributions, all the parameters, including (α_k, β_k, F_k) ($k = 1, \dots, K$) and γ , are identifiable.

Data and likelihood

Data: $\{Y_{ik}, \Delta_{ik}, X_i : i = 1, \dots, n \text{ and } k = 1, \dots, K\}$

- $Y_{ik} = \min(T_{ik}, C_{ik})$
- $\Delta_{ik} = I(T_{ik} \leq C_{ik})$
- C_{ik} : censoring time for the k th event type of the i th subject

Independent censoring assumption: $C_{ik} \perp\!\!\!\perp (T_{ik}, \xi_i)$ given X_i

Likelihood function:

$$L_n(\alpha, \beta, \gamma, F) = \prod_{i=1}^n \prod_{k=1}^K \left[G'_k \{F_k(Y_{ik}) e^{\alpha_k + \beta_k^T X_i}\} F'_k(Y_{ik}) e^{\alpha_k + \beta_k^T X_i} \right]^{\Delta_{ik}} \\ \times \int_{\xi_i} \xi_i^{\sum_{k=1}^K \Delta_{ik}} \exp \left[-\xi_i \sum_{k=1}^K G_k \{F_k(Y_{ik}) e^{\alpha_k + \beta_k^T X_i}\} \right] g(\xi_i; \gamma) d\xi_i,$$

where $g(\xi; \gamma)$ is the density of $\text{Gamma}(\gamma^{-1}, \gamma)$.

NPMLE

We treat F_k as a discrete distribution function with positive jumps at all Y_{ik} with $\Delta_{ik} = 1$.

Then the log-likelihood function is

$$\begin{aligned} \ell_n(\alpha, \beta, \gamma, F) = & \sum_{i=1}^n \sum_{k=1}^K \Delta_{ik} \left[\log G'_k \left\{ F_k(Y_{ik}) e^{\alpha_k + \beta_k^T X_i} \right\} + \log F_k \{Y_{ik}\} + \alpha_k + \beta_k^T X_i \right] \\ & + \sum_{i=1}^n \log \int_{\xi_i} \xi_i \sum_{k=1}^K \Delta_{ik} \exp \left(-\xi_i \left[\sum_{k=1}^K G_k \left\{ F_k(Y_{ik}) e^{\alpha_k + \beta_k^T X_i} \right\} \right] \right) g(\xi_i; \gamma) d\xi_i \end{aligned}$$

We maximize the log-likelihood over α_k , β_k , γ , and the jump sizes of F_k , under the constraint that the sum of all jumps of F_k equals 1.

EM algorithm

The maximization can be solved via an EM algorithm, with gamma frailties ξ_i treated as missing data.

The complete-data log-likelihood function is

$$\begin{aligned} \sum_{i=1}^n \sum_{k=1}^K \Delta_{ik} \left[\log G'_k \left\{ F_k(Y_{ik}) e^{\alpha_k + \beta_k^T X_i} \right\} + \log F_k \{ Y_{ik} \} + \alpha_k + \beta_k^T X_i + \log \xi_i \right] \\ - \sum_{i=1}^n \xi_i \sum_{k=1}^K G_k \left\{ F_k(Y_{ik}) e^{\alpha_k + \beta_k^T X_i} \right\} + \sum_{i=1}^n \log g(\xi_i; \gamma) \end{aligned}$$

E-step

In the E-step, we evaluate the conditional expectation of some function $H(\xi_i)$ given the observed data.

The conditional density of ξ_i given the observed data is proportional to

$$\xi_i^{\sum_{k=1}^K \Delta_{ik}} \exp \left[-\xi_i \sum_{k=1}^K G_k \{ F_k(Y_{ik}) e^{\alpha_k + \beta_k^T X_i} \} \right] g(\xi_i; \gamma) \\ \sim \text{Gamma} \left(\gamma^{-1} + \sum_{k=1}^K \Delta_{ik}, \left[\gamma^{-1} + \sum_{k=1}^K G_k \{ F_k(Y_{ik}) e^{\alpha_k + \beta_k^T X_i} \} \right]^{-1} \right)$$

The integral over ξ_i can be calculated analytically or by a Laplace approximation.

M-step

Notation:

- $t_{1k} < t_{2k} < \dots < t_{m_k, k}$: sorted sequence of all Y_{ik} with $\Delta_{ik} = 1$
- $f_{lk} = F_k\{t_{lk}\}$, for $k = 1, \dots, K$ and $l = 1, \dots, m_k$

In the M-step, we maximize the following objective function:

$$\begin{aligned} M(\alpha, \beta, \gamma, F) = & \sum_{i=1}^n \sum_{k=1}^K \Delta_{ik} \left[\log G'_k \left\{ \sum_{l: t_{lk} \leq Y_{ik}} f_{lk} (Y_{ik}) e^{\alpha_k + \beta_k^\top X_i} \right\} + \log \sum_{l: t_{lk} \leq Y_{ik}} f_{lk} \right. \\ & \left. + \alpha_k + \beta_k^\top X_i + \hat{E}(\log \xi_i) \right] - \sum_{i=1}^n \hat{E}(\xi_i) \sum_{k=1}^K G_k \left\{ \sum_{l: t_{lk} \leq Y_{ik}} f_{lk} e^{\alpha_k + \beta_k^\top X_i} \right\} \\ & - n \log \gamma^{1/\gamma} \Gamma(\gamma^{-1}) + (\gamma^{-1} - 1) \sum_{i=1}^n \hat{E}(\log \xi_i) - \gamma^{-1} \sum_{i=1}^n \hat{E}(\xi_i) \end{aligned}$$

under the constraint $\sum_{l=1}^{m_k} f_{lk} = 1$, for $k = 1, \dots, K$.

M-step (cont.)

The score equation for f_{lk} is

$$\begin{aligned} \frac{1}{f_{lk}} = & - \sum_{i=1}^n I(Y_{ik} \geq t_{lk}) \Delta_{ik} \frac{G''_k \left\{ F_k(Y_{ik}) e^{\alpha_k + \beta_k^T X_i} \right\}}{G'_k \left\{ F_k(Y_{ik}) e^{\alpha_k + \beta_k^T X_i} \right\}} e^{\alpha_k + \beta_k^T X_i} \\ & + \sum_{i=1}^n I(Y_{ik} \geq t_{lk}) \hat{E}(\xi_i) G'_k \left\{ F_k(Y_{ik}) e^{\alpha_k + \beta_k^T X_i} \right\} e^{\alpha_k + \beta_k^T X_i} + \mu_k, \end{aligned}$$

where μ_k is the Lagrange multiplier.

This yields a recursive formula

$$\begin{aligned} \frac{1}{f_{l+1,k}} = & \frac{1}{f_{lk}} + \sum_{i=1}^n I(t_{lk} \leq Y_{ik} < t_{l+1,k}) \Delta_{ik} \frac{G''_k \left\{ F_k(Y_{ik}) e^{\alpha_k + \beta_k^T X_i} \right\}}{G'_k \left\{ F_k(Y_{ik}) e^{\alpha_k + \beta_k^T X_i} \right\}} e^{\alpha_k + \beta_k^T X_i} \\ & - \sum_{i=1}^n I(t_{lk} \leq Y_{ik} < t_{l+1,k}) \hat{E}(\xi_i) G'_k \left\{ F_k(Y_{ik}) e^{\alpha_k + \beta_k^T X_i} \right\} e^{\alpha_k + \beta_k^T X_i} \end{aligned}$$

M-step (cont.)

Similar to Zeng & Lin (2007, JASA), we can then treat $(\alpha_k, \beta_k, f_{1k})$ ($k = 1, \dots, K$) and γ as the parameters to be updated in the M-step, since all other f_{lk} can be expressed as a function of these parameters.

This way, the maximization is carried out over only a small set of parameters, such that the EM algorithm is immune to the high-dimensional parameters in NPMLE.

M-step (cont.)

We can update $(\alpha_k, \beta_k, f_{1k})$ ($k = 1, \dots, K$) and γ via the one-step Newton-Raphson method. The equations to be solved are

$$0 = \sum_{i=1}^n \Delta_{ik} \left[\frac{G_k'' \left\{ F_k(Y_{ik}) e^{\alpha_k + \beta_k^T X_i} \right\}}{G_k' \left\{ F_k(Y_{ik}) e^{\alpha_k + \beta_k^T X_i} \right\}} F_k(Y_{ik}) e^{\alpha_k + \beta_k^T X_i} + 1 \right] (1, X_i^T)^T \\ - \sum_{i=1}^n \hat{E}(\xi_i) G_k' \left\{ F_k(Y_{ik}) e^{\alpha_k + \beta_k^T X_i} \right\} F_k(Y_{ik}) e^{\alpha_k + \beta_k^T X_i} (1, X_i^T)^T, \\ \sum_{l=1}^{m_k} f_{lk} = 1,$$

for $k = 1, \dots, K$, and

$$\frac{n}{\gamma^2} \log \gamma - \frac{n}{\gamma^2} + n \frac{\Gamma'(\gamma^{-1})}{\gamma^2 \Gamma(\gamma^{-1})} - \frac{1}{\gamma^2} \sum_{i=1}^n \hat{E}(\log \xi_i) + \frac{1}{\gamma^2} \sum_{i=1}^n \hat{E}(\xi_i) = 0.$$

Note that f_{lk} is now a function of $(\alpha_k, \beta_k, f_{1k})$, and the derivatives can be calculated based on the recursive formula.

Boundary issue

- One limitation of this EM algorithm is that the estimate of γ must be positive.
- However, when $\gamma = 0$ (i.e., no correlation among all event types), the MLE of γ can be 0 or even negative. The EM algorithm is not applicable due to an improper density of ξ_i .
- In that case, we estimate the other parameters using the same EM algorithm while fixing $\gamma = 0$ and $\hat{E}(\xi_i) = 1$.
- We then compare the observed-data likelihoods with and without the constraint $\gamma = 0$. The estimates with a larger observed-data likelihood will be treated as the final estimates.

Asymptotic properties

Consistency:

$$\sum_{k=1}^K \left(|\hat{\alpha}_k - \alpha_{0k}| + |\hat{\beta}_k - \beta_{0k}| \right) + |\hat{\gamma} - \gamma_0| + \sum_{k=1}^K \sup_{t \in [0, \tau]} |\hat{F}_k - F_{0k}| \xrightarrow{a.s.} 0$$

Asymptotic normality: $\sqrt{n}(\hat{\beta}_k - \beta_{0k}, \hat{\gamma} - \gamma_0, \hat{\Lambda}_k - \Lambda_{0k})_{k=1, \dots, K}$ converges weakly to a mean-zero Gaussian process.

Semiparametric efficiency: The limiting covariances of $\hat{\beta}_k$ ($k = 1, \dots, K$) and $\hat{\gamma}$ attains the semiparametric efficiency bound.

The limiting covariance for $(\hat{\alpha}_k, \hat{\beta}_k, \hat{F}_k)$ ($k = 1, \dots, K$) and $\hat{\gamma}$ can be consistently estimated based on the inverse of the observed information matrix (treating the jump sizes of F_k as usual parameters) and the delta method.

Concluding remarks

- All these papers are rediscussed in Zeng & Lin (2007)⁴. Their likelihood functions can be written in a generic form

$$L_n(\theta, \mathcal{A}) = \prod_{i=1}^n \prod_{k=1}^K \prod_{l=1}^{n_{ik}} \prod_{t \leq \tau} \lambda_k(t)^{\mathbf{d}N_{ikl}(t)} \psi(\mathcal{O}_i; \theta, \mathcal{A})$$

- A general asymptotic theory has been established in Zeng & Lin (2010)⁵.
- To prove the asymptotic properties for each specific problem, we only need to check the regularity conditions of the general theory.

⁴ Zeng, D., & Lin, D. Y. (2007). Maximum likelihood estimation in semiparametric regression models with censored data. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 69(4), 507-564

⁵ Zeng, D., & Lin, D. Y. (2010). A general asymptotic theory for maximum likelihood estimation in semiparametric regression models with censored data. *Statistica Sinica*, 20(2), 871.